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To Prof. Louis H. Kauffman, Editor, *Series on Knots and Everything*
World Scientific Publishing | kauffman@uic.edu

July 18, 2026

Dear Professor Kauffman,

I should like to propose a monograph for the *Series on Knots and Everything*: ***Real Quantum Mechanics: A Multiphase 3D Continuum Formulation, Realised through Mind–AI Cooperation*** — a complete 339-page draft, Volume VI of the *Applied Mathematics: Body and Soul* programme (Volumes I–IV published by Springer, 2003–2009).

The book. The book develops a reformulation of quantum mechanics, *RealQM*, that returns to the real-three-dimensional-space picture Schrödinger originally envisaged in 1926, before the wave function was moved onto $3N$ -dimensional configuration space. The N -electron wave function is a sum of one-electron wave functions on non-overlapping spatial domains separated by free (Bernoulli) boundaries; ground states minimise the ordinary Coulomb energy. There is no configuration space, no probabilistic interpretation, and no antisymmetry postulate — Pauli exclusion is realised geometrically as spatial non-overlap. The formulation is parameter-free at the atomic level and is validated across chemistry (atoms, covalent bonding, hydrides, reactions, condensed phases), extended to a one-paragraph proof of stability of matter and, in its most recent development, to the atomic nucleus itself. There the same Coulomb variational principle — with protons and electrons simply exchanging their inner and outer roles — reproduces nuclear binding energies across the periodic table: the alpha particle at 103% of experiment and the binding-per-nucleon curve, saturation and all, at 107–108%, from a single scale fixed on the deuteron and with no strong force. The nuclear chapters carry this into a Coulomb reading of fusion (a topological rearrangement of proton and electron domains, including a neutrino-free route to solar hydrogen burning) and into a direct comparison with quantum chromodynamics, which after fifty-three years still yields no parameter-free prediction of any nuclear binding energy. This nuclear material has grown into a small suite of companion papers now under journal submission, and forms the most striking part of the book.

Why this series. The book sits naturally in the *Knots and Everything* tradition of foundational, cross-disciplinary mathematical physics that the mainstream journals and the large physics imprints tend to gate out. It is a constructive, computational reformulation in the spirit of the series' established titles, and it makes contact with topology (the free-boundary partition of space into electron territories), variational analysis, and the philosophy of physics. It is, frankly, a book that the conventional venues decline on editorial-policy rather than technical grounds — which is precisely why a series founded to keep such work in print is its right home.

What is distinctive. Two features set the book apart from a conventional monograph. *First*, it is paired with a complete, runnable implementation: roughly 8000 lines of JavaScript and WebGPU compute shaders that execute every quantitative claim interactively in a web browser on a consumer laptop, curated as a public Gallery. The book and the Gallery are designed to be read together; the reader can change a parameter and watch the chemistry respond. *Second*, the implementation and a substantial part of the manuscript were produced through extended collaboration with an AI assistant (Claude, Anthropic), and the book documents that collaboration as part of its contribution — a worked, honest account of a new working method for mathematical science that will interest a readership well beyond quantum chemistry.

Audience and credentials. The intended readership is graduate students and researchers in applied mathematics, computational and theoretical chemistry, the foundations of physics, and scientific computing, as well as readers following the human–AI-collaboration question. The *Body and Soul* series (with Johan Hoffman and the FEniCS finite-element project) has a two-decade track record of pairing mathematics with runnable computation; this volume carries that programme to atomic-scale matter. (On a personal note relevant to the series: I spent 1974–75 as a Dickson Instructor in the Mathematics Department at the University of Chicago.)

The manuscript is complete and available immediately as a PDF, together with the public code repository and Gallery. I should be glad to send the full draft and to discuss the proposal further, and would welcome the series’ technical review.

With my respectful regards,

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